

SentinelVision: Land Type Classification

Using Sentinel-2 Satellite Images (Egypt)

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Data Science Track

Graduation Project Discussion
July 2026

Presentation Outline

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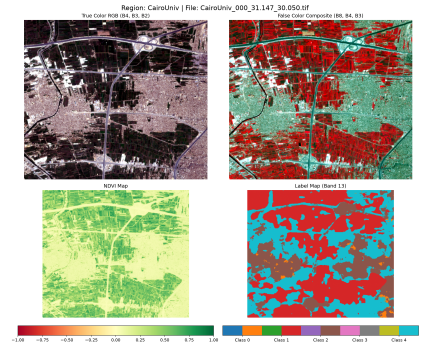
Project Overview & Objectives

Project Motivation

- Satellite imagery is crucial for monitoring rapid changes in urban development, agriculture, and water resources in Egypt.
- Sentinel-2 provides free, high-resolution (up to 10m) multispectral imagery across 12 bands.

Core Objectives

- Develop an end-to-end data science pipeline.
- Design a Deep Neural Network (U-Net) to segment land cover types.
- Deploy the trained model via a fast, interactive web dashboard.



Cairo University Area Composite (RGB)

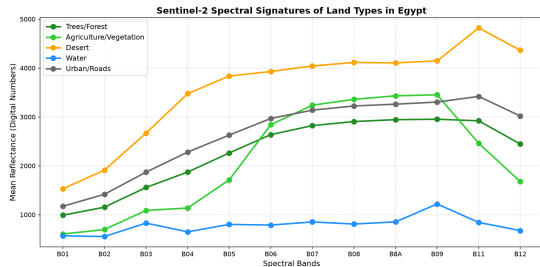
Egypt Sentinel-2 Dataset & Class Mapping

The Dataset Profile

- **Volume:** 719 multi-spectral GeoTIFF files (.tif).
- **Dimensions:** 12 bands $\times$ 256 $\times$ 256 pixels.
- **Labels:** Ground truth masks in Band 13.

Class Frequencies & Semantics

- **0: Trees / Forest** (0.61%) - rare canopy vegetation.
- **1: Agriculture** (22.22%) - Nile Valley / Delta.
- **2: Desert / Soil** (51.80%) - dominant class.
- **3: Water Bodies** (8.37%) - Nile river.



Mean Spectral Signatures per Class

Feature Engineering: Band Selection & NDVI

Normalized Difference Vegetation Index (NDVI)

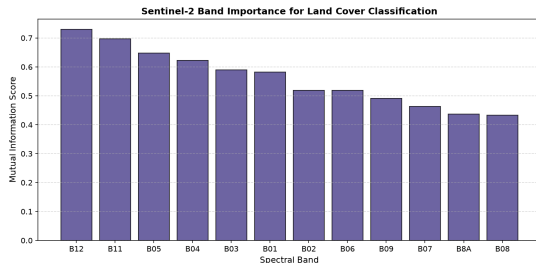
Calculated dynamically using NIR (B8) and Red (B4) bands:

$$\text{NDVI} = \frac{\text{B08} - \text{B04}}{\text{B08} + \text{B04}}$$

Highly discriminative for separating agriculture (NDVI ≈ 0.51) from desert (NDVI ≈ 0.08) and water (NDVI ≈ 0.01).

Mutual Information (MI) Insights

- Rank bands based on information gain.
- Top Features:** NIR (B8), SWIR-1 (B11),



Mutual Information Scores per Band

Principal Component Analysis (PCA)

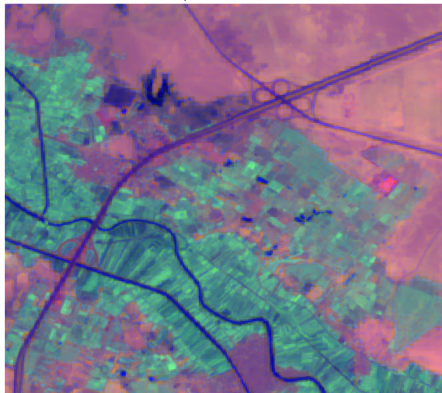
Dimensionality Reduction

- Standardized all 12 spectral bands.
- Projected high-dimensional data onto orthogonal directions of maximum variance.
- **Top 3 PCs** capture over 94% of total variance.

Visualizing PCA (PC1 $\rightarrow$ R, PC2 $\rightarrow$ G, PC3 $\rightarrow$ B)

- Enhances environmental contrasts.
- Separates dry sand, wet agricultural soil, salt basins, and urban infrastructure.

PCA Color Composite (PC1, PC2, PC3)
Total Explained Variance: 98.26%

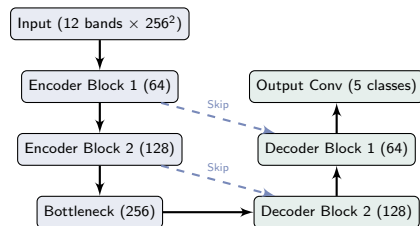


False Color PCA Composite (PCs 1, 2, 3)

U-Net Deep Learning Architecture

Model Architecture

- **Input:** 12 channels (multispectral bands) at 256×256 pixels.
- **Contracting Encoder path:** Extract hierarchical spatial context via double convolutions, batch normalization, and max pooling.
- **Expanding Decoder path:** Reconstruct resolution via up-convolutions.
- **Skip Connections:** Concatenate high-resolution features from encoder to decoder to preserve sharp boundary details.
- **Output:** 5-channel logit map (pixel-level class probabilities).



U-Net Information Flow

Handling Severe Class Imbalance

Because desert covers 52% of pixels and forest covers 0.6%, we calculate **inverse-frequency weights** to scale the cross-entropy loss. Minority classes are penalized heavily:

$$w_c = \frac{1}{\text{frequency}_c} \quad \text{scaled to sum to 1.0}$$

Training Hyperparameters

- **Optimiser:** AdamW (weight decay = 1e-4) for robust regularization.
- **Scheduler:** Cosine Annealing Learning Rate (starts at 1e-3, decays to 1e-6).
- **Epochs:** 10 (fully trained on CUDA GPU).
- **Splits:** Region-stratified: 70% Train (503 images), 15% Val (108), 15% Test (108).

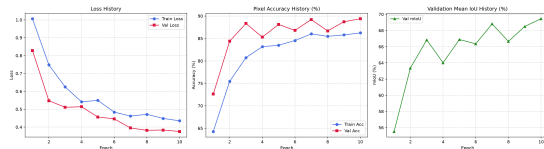
Training Curves & Performance Metrics

Performance Metrics Achieved

- **Overall Pixel Accuracy: 89.36%**
- **Mean Intersection over Union (mIoU): 69.46%**

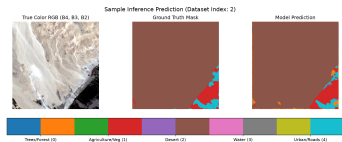
Class-specific IoU Scores

- **Water Bodies: 92.68%** (Excellent)
- **Desert / Bare Soil: 87.03%** (Strong)
- **Agriculture / Vegetation: 86.75%** (Strong)
- **Urban / Roads: 65.11%** (Moderate)
- **Trees / Forest: 15.74%** (Challenging due to scarcity)

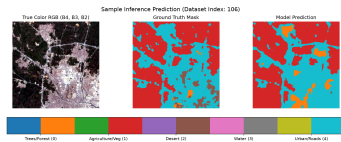


Loss & mIoU convergence over 10 Epochs

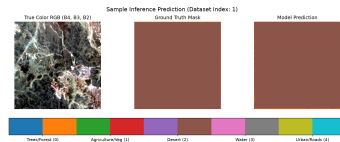
Qualitative Results: Semantic Segmentation Maps



Nile Delta Agricultural Grid



Suez Canal (Water vs Desert)

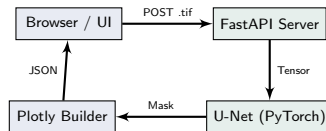


Cairo Urban Expansion

Deployment: SentinelVision Web Application

System Architecture

- **Backend:** FastAPI (Python) server.
- **Frontend:** Single-page HTML5 dashboard styled with Tailwind CSS, leveraging Plotly.js for interactive charts.
- **Inference:** GeoTIFF upload → memory stream → U-Net evaluation → Plotly charts generated on the fly.



GUI System Information Flow

Plotly Charts Generated

- **True Color composite:** stretched Red/Green/Blue bands.
- **Classification Map:** Heatmap aligned with class colors.

Conclusion & Future Scope

Summary of Achievements

- Successfully built and trained a multispectral (12-band) U-Net segmentation model achieving **89.36% accuracy** and **69.46% mIoU**.
- Addressed class imbalance using normalized inverse-frequency weighted cross-entropy.
- Deployed the model on an interactive web dashboard with zero disk-write footprint.

Future Enhancements

- Use transformer-based architectures (e.g. SegFormer) for improved road/urban boundary accuracy.
- Expand the dataset to include multitemporal seasonal imagery of Egypt.
- Deploy on cloud instances like Google Cloud Run or AWS EC2 for public access.

Thank You!

Questions & Discussion